Relational $\mathbb{Z}_4$ Causets:

Algebraic Foundations

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Abstract

A companion paper argues that the cyclic group $\mathbb{Z}_4$ is the minimal algebra (within stated constraints) carrying the relational concepts of identity, opposition, and irreducible otherness. Here we develop the mathematical framework that follows: relational $\mathbb{Z}_4$ causets (directed graphs with $\mathbb{Z}_4$ phase labels), a node conservation law entailed by algebraic consistency, and a two-layer architecture in which the directed graph provides causal structure while the derived holonomy landscape provides constraint structure. We prove that a structural split is inherent in the algebra: Layer 2 observables are direction-insensitive. We analyse initial conditions under two interpretations of the origin—the *participatory void* and the *absolute void*—and show that under the more austere interpretation, the otherness modes ($\phi = \pm i$) reach self-propagating dynamics in fewer steps than the opposition mode ($\phi = -1$).

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1 Introduction

A companion paper (*What Must Exist*) argues that any minimally expressive relational structure requires three primitive concepts—identity, opposition, and irreducible otherness—and that these, together with closed and associative composition, uniquely determine the cyclic group $\mathbb{Z}_4$.

This paper takes $\mathbb{Z}_4$ as given and develops the mathematical structures that arise when $\mathbb{Z}_4$ is used to label directed relations constrained by local algebraic consistency. The central objects are relational $\mathbb{Z}_4$ causets: finite directed graphs with $\mathbb{Z}_4$ phase labels on edges and a conservation law at nodes. We establish the basic definitions, identify the two-layer architecture, prove several structural results, and apply them to the question of initial conditions.

We proceed as follows. Section 2 fixes notation for $\mathbb{Z}_4$. Section 3 defines relational $\mathbb{Z}_4$ causets. Section 4 introduces holonomy and frustration. Section 5 derives the conservation law. Section 6 identifies the two-layer architecture and its structural split. Section 8 states the postulates and describes the dynamics. Sections 9–13 ask what comes first. Section 15 surveys open questions.

2 The group $\mathbb{Z}_4$

Definition 2.1. $\mathbb{Z}_4 = \{1, i, -1, -i\}$ is the cyclic group of order four under multiplication, realised as the fourth roots of unity. The identity is 1. The unique element of order 2 is -1. The elements i and $-i$ have order 4 and are mutual inverses: $i \cdot (-i) = 1$.

The multiplication table is:

$\cdot$	1	i	-1	$-i$
1	1	i	-1	$-i$
i	i	-1	$-i$	1
-1	-1	$-i$	1	i
$-i$	$-i$	1	i	-1

Definition 2.2 (Cayley metric)**.** The *Cayley graph distance* on $\mathbb{Z}_4$ is

$$d(h, 1) = \begin{cases} 0 & h = 1 \\ 1 & h = \pm i \\ 2 & h = -1 \end{cases} \tag{1}$$

the minimum number of generator steps from h to the identity.

Proposition 2.3. *$d(h, 1) = d(h^{-1}, 1)$ for all $h \in \mathbb{Z}_4$.*

Proof. The cases $h = 1$ and $h = -1$ are immediate (-1 is self-inverse). For $h = i$: $i^{-1} = -i$ and $d(i, 1) = d(-i, 1) = 1$. □

Remark 2.4. Following the companion paper: 1 represents *identity*, -1 represents *opposition* (self-inverse), and $i, -i$ represent *otherness* and its conjugate (not self-inverse; $i^{-1} = -i \neq i$). The structural distinction between self-inverse and non-self-inverse elements will have algebraic consequences in what follows.

3 Relational $\mathbb{Z}_4$ causets

Definition 3.1. A *relational* $\mathbb{Z}_4$ *causet* is a triple $G = (V, E, \phi)$ where V is a finite set of nodes, $E \subseteq V \times V$ is a set of directed edges (with $u \to v$ denoting $(u, v) \in E$), and $\phi : E \to \mathbb{Z}_4$ assigns a phase label to each edge.

Definition 3.2. For $v \in V$: $\mathrm{in}(v) = \{e \in E : e = (u, v)\}$ is the set of incoming edges; $\mathrm{out}(v) = \{e \in E : e = (v, w)\}$ the set of outgoing edges.

The edge $u \to v$ means v *is derived from* u: the arrow points from ground to product. Direction reversal maps $\phi(u \to v) \mapsto \phi(u \to v)^{-1}$.

Remark 3.3 (Nodes are derived). V is not ontologically fundamental. A node is the equivalence class of edge-endpoints identified by local relational context. Two nodes with identical edge-bundles are indiscernible. An isolated node has no operational meaning.

3.1 Ontological commitments

We are explicit about the ontological hierarchy:

1. *Events* (edge-creations) are fundamental: the primitive acts of the theory.
2. *Edges* are the relations established by events. Each edge is directed and carries a $\mathbb{Z}_4$ phase label.
3. *Nodes* are derived: a node is the meeting-point of its incident edges, with no independent existence.

A relational $\mathbb{Z}_4$ causet is not a static graph. It is the accumulated record of events—a history of acts of becoming. The graph at any stage is a snapshot of the history so far.

3.2 Events and nodes

Each edge-creation is an *event*.

Definition 3.4. An *event* is the creation of a directed edge $e_k = (s \to t,\ \phi_k)$. If t (or s) is new, the event co-constitutes that node.

Events are partially ordered by their causal dependencies. Nodes are derived, persistent, and accumulative: a node participates in events at different stages.

The partial order is over events, not nodes. A directed cycle in the graph is a cycle among nodes, not among events; the event ordering is always acyclic, even when the node graph contains directed cycles.

4 Holonomy and frustration

Definition 4.1 (Face)**.** A *face* of G is a directed cycle $C = (v_0, v_1, \ldots, v_{k-1}, v_0)$ with $k \geq 3$ distinct vertices that is *induced*: no edge in E connects two non-adjacent vertices of C. Faces are considered up to cyclic permutation of vertices.

Definition 4.2 (Holonomy)**.** The *holonomy* of a face $C = (v_0, \ldots, v_{k-1}, v_0)$ is

$$\mathrm{Hol}(C) = \prod_{j=0}^{k-1} \tilde{\phi}(v_j, v_{j+1}) \tag{2}$$

where indices are mod k and

$$\tilde{\phi}(u, v) = \begin{cases} \phi(u \to v) & \text{if } u \to v \in E, \\ \phi(v \to u)^{-1} & \text{if only } v \to u \in E. \end{cases}$$

Reversing the traversal orientation gives $\mathrm{Hol}(C)^{-1}$.

Definition 4.3 (Frustration)**.** The *frustration* of a face is $\mathcal{F}(C) = d(\mathrm{Hol}(C), 1)$. The *total frustration* of a labelling is $F = \sum_C \mathcal{F}(C)$ over all faces.

Proposition 4.4 (Frustration is undirected)**.** *$\mathcal{F}(C)$ is independent of traversal orientation.*

Proof. Reversing orientation gives holonomy h^{-1}. By definition 2.3, $d(h^{-1}, 1) = d(h, 1)$. □

5 The conservation law

Definition 5.1 (Conservation)**.** A node v *satisfies conservation* if

$$\prod_{e \in \mathrm{in}(v)} \phi(e) \;=\; \prod_{e \in \mathrm{out}(v)} \phi(e) \tag{3}$$

where the empty product is 1. A graph is *conservation-consistent* if every node satisfies conservation.

Definition 5.2 (Charge)**.** The *charge* at a node is

$$q(v) = \left(\prod_{e \in \mathrm{in}(v)} \phi(e) \right) \cdot \left(\prod_{e \in \mathrm{out}(v)} \phi(e) \right)^{-1}. \tag{4}$$

Conservation holds at v iff $q(v) = 1$.

Proposition 5.3 (Uniqueness of the conservation law)**.** *Let R be a local balance condition at a node v in a relational $\mathbb{Z}_4$ causet, satisfying:*

(i) Locality*: R depends only on edges incident to v.*

(ii) Group compatibility*: R is expressible in terms of the $\mathbb{Z}_4$ multiplication, with no external parameters.*

(iii) Node symmetry*: R does not distinguish between nodes—it takes the same form at every v.*

Then, under the assumption that no element of $\mathbb{Z}_4$ is distinguished independently of the relational structure, R is equivalent to $\prod_{\mathrm{in}} \phi(e) = \prod_{\mathrm{out}} \phi(e)$.

Proof. By (i) and (ii), R relates the incoming and outgoing phase products through the group operation. The most general such relation is $\prod_{\mathrm{in}} \phi = g \cdot \prod_{\mathrm{out}} \phi$ for some fixed $g \in \mathbb{Z}_4$. By (iii), g must be the same at every node. But $\mathbb{Z}_4$ has no canonical non-identity element: there is no distinguished choice of $g \neq 1$ available without additional structure. Any $g \neq 1$ would impose an asymmetry between nodes that is not grounded in the ontology. Therefore $g = 1$, giving (3).

At the philosophical level, this condition corresponds to the requirement of local non-contradiction developed in the companion paper; here it appears in its simplest algebraic form. □

Proposition 5.4 (Global charge conservation)**.** $\prod_{v \in V} q(v) = 1$.

Proof. Each edge $e = (u \to v,\, \phi)$ contributes ϕ^{-1} to $q(u)$ (via the outgoing product) and ϕ to $q(v)$ (via the incoming product). The global product telescopes to 1. □

6 The two-layer architecture

Definition 6.1 (Layer 1)**.** The *causet layer* (Layer 1) is the directed graph (V, E, ϕ) itself: the append-only, directed structure of edges and their $\mathbb{Z}_4$ labels.

Definition 6.2 (Layer 2)**.** The *holonomy landscape* (Layer 2) is the collection of all faces, their holonomies, their frustrations, and the charge distribution. Layer 2 is derived from Layer 1: given (V, E, ϕ), every Layer 2 quantity is computable.

Theorem 6.3 (Direction-insensitivity of Layer 2)**.** *Layer 1 is directed. Layer 2 observables are direction-insensitive.*

Proof. Layer 1 is directed by definition: $u \to v \neq v \to u$.

For Layer 2: frustration is direction-insensitive (definition 4.4). Charge is a scalar at a node (definition 5.2). Since frustration and charge are the observable quantities of Layer 2, and both are insensitive to edge direction, Layer 2 observables carry no directional information. □

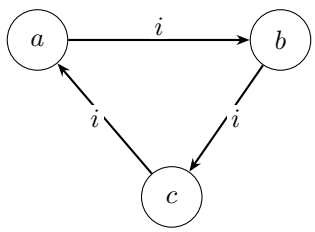
Remark 6.4 (Interpretation). Definition 6.3 establishes a structural distinction: Layer 1 is directed; Layer 2 is not. We do not claim that spacetime structure has been derived; rather, the formal result admits an interpretation suggestive of such a structure—Layer 1 as causal/temporal, Layer 2 as constraint/relational, possibly underlying spatial geometry. We adopt this as a heuristic guide in what follows, while noting that a rigorous identification with spacetime requires showing that Layer 2 structure organises into spatial dimensions (the open problem of dimensional stability).

7 Worked examples

We illustrate the two-layer architecture with two small causets.

Example 7.1 (Frustrated triangle)**.** Consider the relational $\mathbb{Z}_4$ causet with three nodes $\{a, b, c\}$ and three edges, all carrying phase i.

Layer 1 (directed graph with $\mathbb{Z}_4$ labels):



Three directed edges, each with phase i. The direction matters: $a \to b$ means b is derived from a. Reversing any edge would conjugate its phase ($i \mapsto -i$).

Layer 2 (derived holonomy landscape):

The graph contains one face: the induced 3-cycle (a, b, c, a).

Holonomy. Traversing in the directed sense:

$$\mathrm{Hol}(a \to b \to c \to a) = i \cdot i \cdot i = i^3 = -i.$$

Traversing in the reverse sense gives $\mathrm{Hol} = (-i)^{-1} = i$. The two values are conjugate (inverses in $\mathbb{Z}_4$).

Frustration. $\mathcal{F} = d(-i, 1) = 1$. Equivalently, $d(i, 1) = 1$: the same value regardless of traversal direction. **Frustration is undirected**—it is a property of the cycle as an unoriented loop, not of its traversal.

Charge. At each node, one edge arrives and one departs, both carrying i:

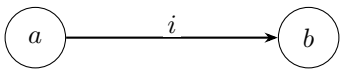
Node	$\prod \phi_{\text{in}}$	$\prod \phi_{\text{out}}$	$q(v)$
a	i	i	1
b	i	i	1
c	i	i	1

All charges vanish. The graph is conservation-consistent.

The structural split in action. Layer 1 carries directed information (the arrows, the distinction between $a \to b$ and $b \to a$). Layer 2 carries undirected information (frustration = 1, charges = 1 at every node). This concretely illustrates definition 6.3: although the underlying graph is directed, the derived frustration is insensitive to orientation.

Remark 7.2. An analogous triangle with $\phi = -1$ (opposition) yields holonomy $(-1)^3 = -1$ and frustration 2, illustrating the distinction between otherness and opposition modes. The difference in frustration—1 for otherness, 2 for opposition—reappears in the bootstrap analysis (section 13).

Example 7.3 (Charged pair: Layer 1 without Layer 2)**.** The minimal graph: a single edge $a \to b$ with phase i.



Layer 1: One directed edge. b is derived from a.
Layer 2: No faces exist ($k \geq 3$ required), so there is no holonomy and hence no frustration; the constraint layer has not yet come into existence. This graph has Layer 1 content but no Layer 2 content: directed structure without constraint structure.

Charge. Conservation fails at both nodes:

Node	$\prod \phi_{\text{in}}$	$\prod \phi_{\text{out}}$	$q(v)$
a	1 (empty)	i	$-i$
b	i	1 (empty)	i

The charges are conjugate: $q(a) \cdot q(b) = (-i)(i) = 1$ (definition 5.4).

This is the initial state of the growth sequence: a charged pair with directed structure but no constraint structure. The frustrated triangle of definition 7.1 is what it grows into—acquiring Layer 2 content when the first face closes.

8 Dynamics

The theory rests on three postulates:

1. Directed edges with $\mathbb{Z}_4$ phase labels are fundamental. Nodes are derived.
2. All dynamics is local.
3. Algebraic consistency: the conservation law (3) holds; new edges must be consistent with existing local phase structure.

The dynamics creates structure through two moves: *sprouting* (extending an edge to a new node) and *bridging* (connecting two existing nodes, creating faces). We adopt the dynamical hypothesis that frustration drives growth: regions of high frustration create pressure for new edges that redistribute it, and bridging is preferred over sprouting because it immediately creates faces, enabling constraint propagation. We adopt this as the minimal dynamics coupling Layer 2 structure back to Layer 1 growth: frustration is the only scalar quantity derived from holonomy that is available at Layer 2 without additional assumptions. This hypothesis is not proved here; it is motivated by the crystallisation dynamics developed in the companion physics paper and is assumed in the dynamical arguments (but not the theorems) that follow.

9 Back to the beginning

We now ask what the minimal structure is from which dynamics can proceed.

Definition 9.1 (Growth sequence)**.** A *growth sequence* is a finite sequence of events $e_1, \ldots, e_n$ starting from the empty graph, where each $e_k = (s_k \to t_k, \phi_k)$ either bridges or sprouts, and $e_1 = (a \to b, \phi)$ co-constitutes both a (*void*, ground-pole) and b (product-pole).

Definition 9.2 (Sprouting tree)**.** The *sprouting tree* of a growth sequence is the subgraph of all sprouting edges: a directed tree rooted at b with the edge $a \to b$ as its root edge from outside the tree.

Definition 9.3 (Self-propagating)**.** A growth sequence is *self-propagating* if the resulting graph contains at least one frustrated face.

Charges alone do not suffice for self-propagation: crystallisation operates through constraint propagation along faces, which requires holonomy structure that a faceless graph lacks.

10 The origin

The first event co-constitutes both nodes of the first edge. Under the causal reading, b is derived from a. Whether a can subsequently appear as a target—whether later structure can claim causal priority over the origin—is an interpretive question with algebraic consequences.

Definition 10.1 (Participatory void)**.** a may appear as the target of edges created after e_1.

Definition 10.2 (Absolute void)**.** a never appears as a target: $t_k \neq a$ for all k.

Both are compatible with the postulates. The *participatory void* allows the origin to be drawn back into the structure it seeds—it becomes part of the spatial fabric. The *absolute void* is permanently upstream: the uncaused ground, outside the world it gives rise to. We develop both.

Remark 10.3. The distinction here is solely about whether the void can be a *target* of later edges. Under both interpretations, the void may serve as a source (emitting one or more edges).

11 Algebraic lemmas

Lemma 11.1 (Phase propagation). *In a conservation-consistent directed path where each intermediate node has exactly one incoming and one outgoing edge, all edges carry the same phase.*

Proof. Conservation at each intermediate node equates incoming and outgoing phase. □

Lemma 11.2 (No nontrivial idempotents). *The only solution to $\phi^2 = \phi$ in $\mathbb{Z}_4$ is $\phi = 1$.*

Proof. $1^2 = 1$; $i^2 = -1 \neq i$; $(-1)^2 = 1 \neq -1$; $(-i)^2 = -1 \neq -i$.
Alternatively: in any group, $\phi^2 = \phi$ implies $\phi = e$. □

Lemma 11.3 (Junction constraint). *A conservation-consistent node with two incoming edges of phase ϕ and one outgoing edge of phase ϕ satisfies $\phi = 1$.*

Proof. $\phi^2 = \phi$. Apply definition 11.2. □

12 Participatory void

Theorem 12.1. *Under the participatory void with $\phi \neq 1$, the minimal self-propagating seed is the triangle $a \to b \to c \to a$: three edges, all phase ϕ, holonomy ϕ^3.*

Proof. Fewer than three edges cannot form a face. Three edges $a \to b$, $b \to c$, $c \to a$: conservation forces all phases to ϕ (definition 11.1). Holonomy $\phi^3 \neq 1$ for $\phi \neq 1$. No other three-edge growth sequence with a participatory void creates a face. □

Remark 12.2. The void-return resolves all charges and does not distinguish Dirac-type ($\phi = \pm i$, holonomy $\mp i$, frustration 1) from Majorana-type ($\phi = -1$, holonomy -1, frustration 2). (We use "Dirac-type" and "Majorana-type" as structural labels chosen by analogy only, reflecting the conjugate-pair vs. self-conjugate distinction in $\mathbb{Z}_4$. No claim is made about correspondence to fermion physics.)

13 Absolute void

Under the absolute void, no face passes through a. The void has no Layer 2 content: it is purely temporal.

Theorem 13.1. *Under the absolute void with single emission $e_1 = (a \to b, \phi)$, $\phi \neq 1$:*

(a) Bridges from internal tree nodes carry phase 1. Bridges from leaves carry phase ϕ and force $\phi = 1$ at internal targets.
(b) Dirac-type *($\phi = \pm i$): minimum 5 edges. Chain $a \to b \to c \to d \to e$, bridge $d \to b$ (phase 1). Triangle $b \to c \to d \to b$, holonomy -1.*
(c) Majorana-type *($\phi = -1$): minimum 6 edges. Chain $a \to b \to c \to d \to e \to f$, bridge $e \to b$ (phase 1). 4-cycle $b \to c \to d \to e \to b$, holonomy -1.*
(d) Dirac-type reaches frustration in fewer edges than Majorana-type.

Proof. Part (a). All tree edges carry ϕ (definition 11.1). A leaf source u with one incoming ϕ and one outgoing bridge ψ: conservation gives $\psi = \phi$. At an internal target: $\phi \cdot \phi = \phi$, so $\phi = 1$ (definition 11.3). An internal source u with one incoming ϕ and one outgoing ϕ (tree child) plus bridge ψ: $\phi = \phi \cdot \psi$, so $\psi = 1$.

Part (b). Triangle from two tree edges and one bridge: holonomy $\phi^2 \cdot 1 = \phi^2$. For $\phi = \pm i$: $\phi^2 = -1$. Both bridge endpoints must be internal (each needs a tree child). Shortest chain: $a \to b \to c \to d \to e$. Bridge $d \to b$. Total: 5. Conservation holds at all interior nodes.

Four edges: chain $a \to b \to c \to d$ plus bridge. $c \to b$ gives a 2-cycle. $d \to b$: leaf-sourced, forces $\phi = 1$. No frustrated face.

Part (c). Triangle holonomy $(-1)^2 = 1$: flat. A 4-cycle with three tree edges and bridge: $(-1)^3 \cdot 1 = -1$. Chain $a \to b \to c \to d \to e \to f$, bridge $e \to b$. Total: 6.

Five edges give at most a triangle (holonomy 1).

Part (d). $5 < 6$. □

Corollary 13.2 (Dirac-type selection)**.** *Under the absolute void, the $\mathbb{Z}_4$ algebra selects the otherness modes ($\phi = \pm i$) over the opposition mode ($\phi = -1$) by economy.*

Corollary 13.3 (Linear tree)**.** *Under the absolute void with $\phi \neq 1$, the sprouting tree is linear (at most one child per node) until bridges are added.*

Proof. Multi-child sprouting from one node with a single incoming phase ϕ: $\phi = \phi^d$ for $d \geq 2$ outgoing tree edges, giving $\phi^{d-1} = 1$. For $d = 2$: $\phi = 1$ (definition 11.2). □

The pre-bridge universe is one-dimensional. The frustration-driven dynamics escapes linearity at the earliest opportunity: bridging creates faces, which sprouting cannot. One dimension is the forced starting point; the dynamics departs from it as soon as a viable bridge exists.

Proposition 13.4 (Balanced emission)**.** *With an absolute void emitting ϕ and ϕ^{-1} (product 1, no void charge), the first frustrated face requires 6 edges.*

Proof. $e_1 = (a \to b, \phi)$, $e_2 = (a \to d, \phi^{-1})$, $e_3 = (b \to c, \phi)$, $e_4 = (d \to c, \phi^{-1})$, $e_5 = (c \to f, 1)$, $e_6 = (f \to b, 1)$. Triangle $b \to c \to f \to b$: holonomy ϕ. □

14 Comparison of interpretations

	Participatory	**Absolute**
First frustration (Dirac-type)	3 edges	5 edges
First frustration (Majorana-type)	3 edges	6 edges
Dirac/Majorana distinguished	No	Yes
Void charge	Resolved	Permanent
Void on a face	Yes	No
Void's Layer 2 role	Spatial	None

The participatory void is more economical (3 vs. 5 edges) but does not select between initial phases. The absolute void is less economical but resolves the Dirac-type/Majorana-type ambiguity: otherness is preferred over opposition by economy. Under the participatory void, the selection must come from subsequent dynamics.

15 Open questions

Dimensional stability. The linear tree escapes one dimension via the first bridge. What equilibrium connectivity does the dynamics select? This is the question of whether 3+1 dimensions emerge.

Void interpretation. Both participatory and absolute voids are compatible with the postulates. Whether the theory favours one, or whether the distinction has observable consequences at large scales, is open.

Economy as dynamics. The Dirac-type seed reaches frustration in fewer edges, but we have shown economy as a structural distinction, not a dynamical law. Whether the crystallisation dynamics preferentially follows the shortest path to frustration is a question for simulation.

Frustration as the dynamical scalar. We have adopted frustration as the quantity mediating between Layer 2 constraints and Layer 1 growth. Whether frustration is the unique scalar capable of playing this role—or whether other Layer 2 quantities (e.g. charge density, face count) could serve as alternative or complementary drivers—remains open.

Continuum limit. Whether the frustration of faces approximates the Einstein–Hilbert action in an appropriate limit is the bridge to general relativity.

Sufficiency of $\mathbb{Z}_4$. Whether $\mathbb{Z}_4$ is rich enough for all of physics, or is a proof-of-concept for a framework requiring a larger gauge group, remains open. The companion paper argues for the conceptual minimality of $\mathbb{Z}_4$; the physical sufficiency is a separate question.